

CAPSTONE · SUPPLY CHAIN ANALYTICS

Late Delivery Risk Prediction

Can we predict which DataCo orders will arrive late – and what should we do about it?

65,752
orders analysed

54.8%
delivered late

0.77
model ROC-AUC

01 · 2 MIN
Purpose & objectives

02 · 2 MIN
Data & exploration

03 · 6 MIN
Five objectives

04 · 3 MIN
Predictive model

05 · 2 MIN
Findings → Q&A

AT A GLANCE

Shipping mode drives late delivery

65,752 orders · Jan 2015 – Jan 2018 · use the slicers to filter the KPIs and the shipping-mode chart

Year: 2015 – 2018

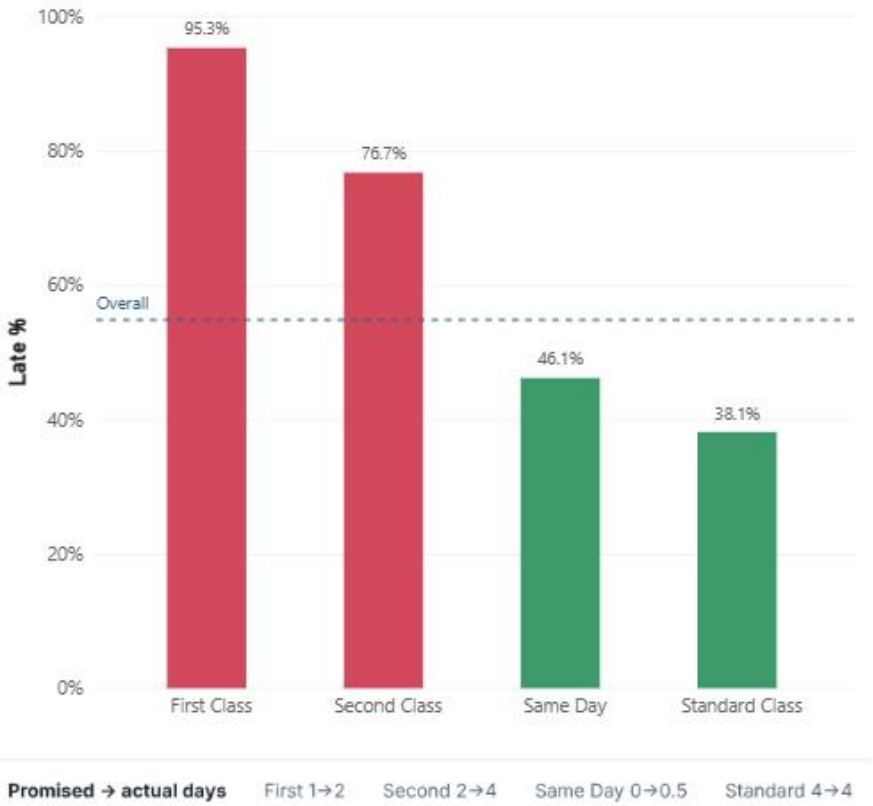
Market: Africa | Europe | LATAM | Pacific Asia | USCA

Late Delivery RISK PREDICTION

Orders	Late %	Promised (days)	Actual (days)	model ROC-AUC
66K	54.8%	2.93	3.50	0.77 · order-time data only

1 Shipping mode is the driver

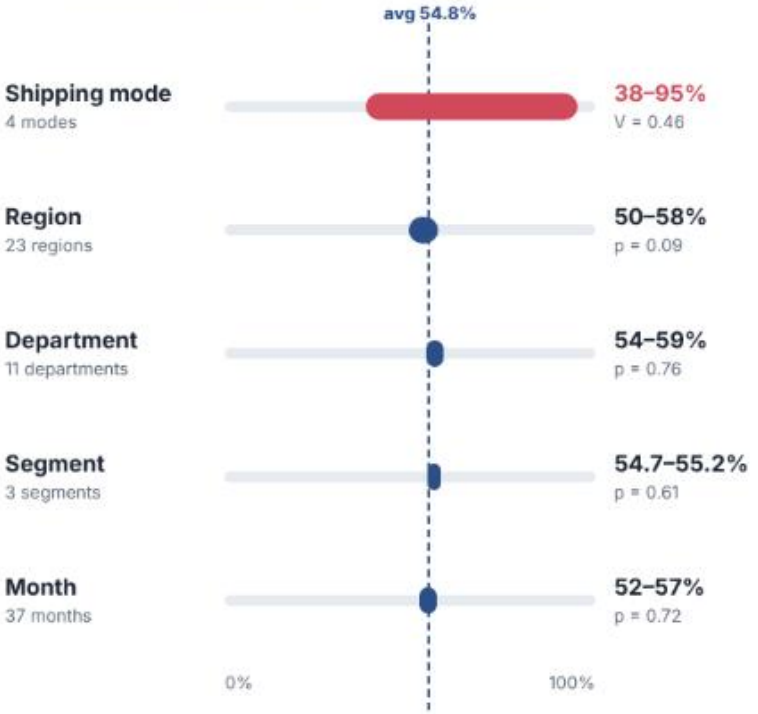
Late % by shipping mode · live, responds to the slicers



Promised → actual days: First 1→2 · Second 2→4 · Same Day 0→0.5 · Standard 4→4

2 Everything else is flat

Lowest to highest late % within each factor · all orders



3 The model confirms it

Gradient Boosting · 13,151 test orders

86% precision — flagged orders truly late

58% recall — late orders caught

What the model relies on

Shipping mode	0.245
Order hour	0.044
Payment type	0.015
All others	≈ 0

Confusion matrix · test set, 13,151 orders

	Predicted late	Predicted on time
Actually late	4,170 caught	3,040 missed
Actually on time	698 false alarms	5,243 correct on time

TAKEAWAY **Fix the promise on fast shipping modes and score orders at checkout;** region, category, segment and timing fixes won't pay off.

Purpose and problem statement

Late delivery hurts customer satisfaction, cost and repeat business

Purpose

1 **Understand**
Where and why late deliveries happen in DataCo's supply chain

2 **Predict**
Build an early-warning model that flags risky orders at order time

3 **Act**
Give operations clear, evidence-based actions

Problem statement

*"How can we analyse and predict whether an order is at risk of late delivery based on **shipping mode**, **geography** and **product category**?"*

SCALE OF THE PROBLEM

54.8% of all orders arrive late

TAKEAWAY **54.8%** of all orders arrive late – a systemic issue, not an edge case.

RESEARCH OBJECTIVES

Five questions about late delivery

Each factor is tested against the **54.8%** overall late rate · 65,752 orders, Jan 2015 – Jan 2018

Late Delivery
RISK PREDICTION

01



OBJECTIVE 1

Shipping mode

"How do shipping modes influence late delivery?"

LOOKS AT

- Shipping mode
- Promised vs actual days

χ^2 test · Cramér's V

Slide 8 →

02



OBJECTIVE 2

Geographic risk

"How does risk vary across regions, countries and cities?"

LOOKS AT

- Market
- Region
- Country
- City

χ^2 test · Cramér's V

Slide 9 →

03



OBJECTIVE 3

Product category

"Are some categories or departments more often late?"

LOOKS AT

- Department
- Category

χ^2 test · Cramér's V

Slide 10 →

04



OBJECTIVE 4

Customer segment

"Does risk differ by segment or customer value?"

LOOKS AT

- Segment
- Customer value tier

χ^2 test · Cramér's V

Slide 11 →

05



OBJECTIVE 5

Temporal trends

"Is late-delivery risk cyclical, improving or worsening?"

LOOKS AT

- Monthly late %
- 3-month trend

χ^2 test · Cramér's V

Slide 12 →

THEN **Predict:** can we flag a late order at checkout, using only what we know when it's placed?

Gradient Boosting · Logistic Regression · Random Forest

Dataset and preparation

DataCo supply chain tables merged into one order-level table

65,752
orders analysed

180,519
order items

2015–2018
Jan 2015 – Jan 2018

54.8%
late rate

Sources

- Orders, Shipping, Geography, Payment Types, Customers
- Order Items → Products → Categories → Departments

Raw tables → **Clean & merge** → **Features** → **Model**

Preparation

- Customer personal data removed
- Mixed date formats parsed; delay = actual − scheduled days
- Order-level features and customer value tier built
- Only Order Zipcode had material missing values

TAKEAWAY The late flag matches “actual > scheduled days” in **97.5%** of orders – a genuine delivery outcome.

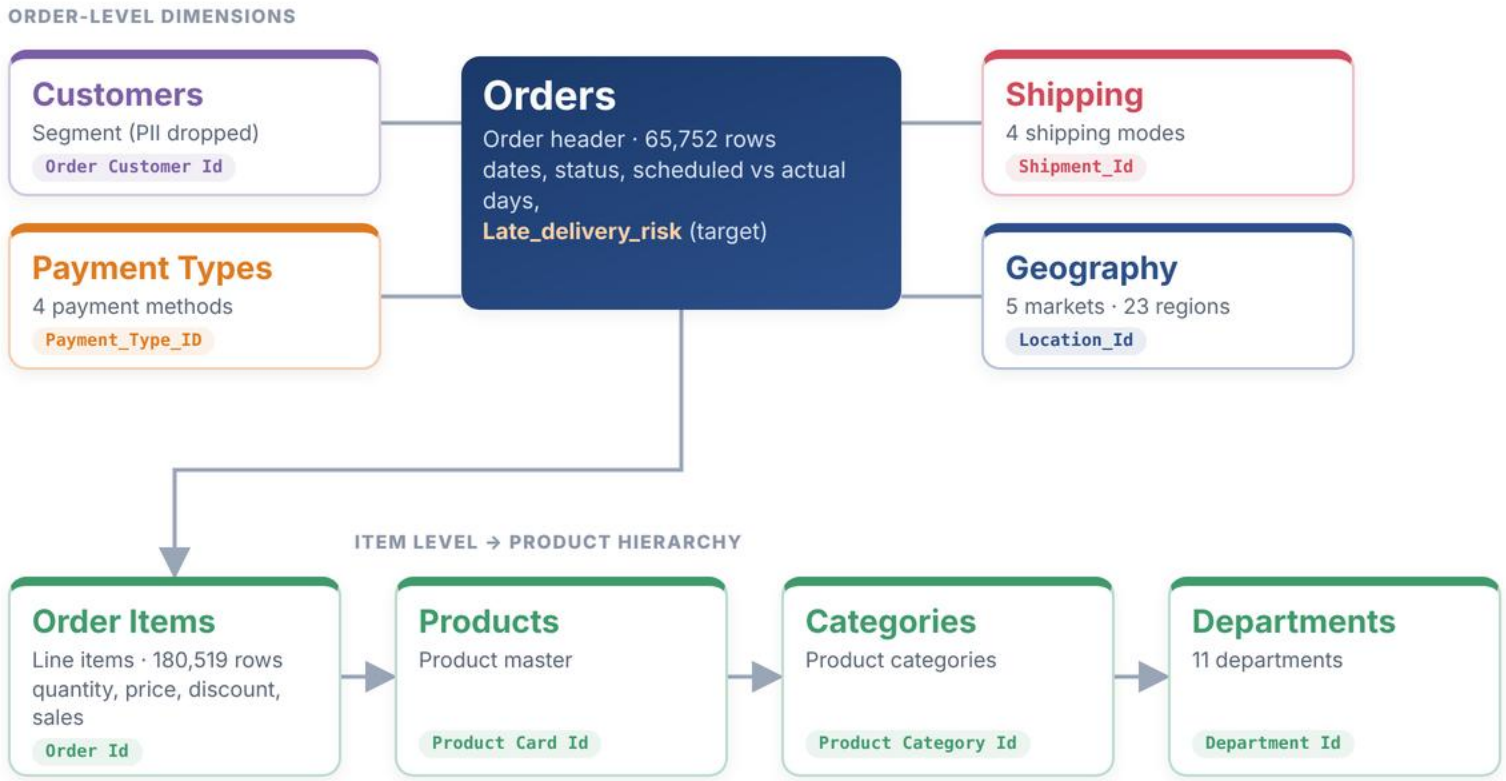
DATA CONTEXT

Where the data comes from

DataCo Supply Chain Analytics · 9 related tables, joined into one row per order

Late Delivery
RISK PREDICTION

DATA MAP · HOW THE 9 TABLES LINK



SOURCE & SCOPE

DataCo Supply Chain Analytics

Public dataset of end-to-end order fulfilment for a global retailer: customers, products, payments, shipping and markets.

65,752 orders	**180,519** order line items
37 months Jan 2015 – Jan 2018	**5 · 23** markets · regions
4 · 4 shipping modes · payment types	**3 · 11** customer segments · departments

Target: `Late_delivery_risk` (1 = late) · **54.8%** of orders are late

PREPARED FOR ANALYSIS

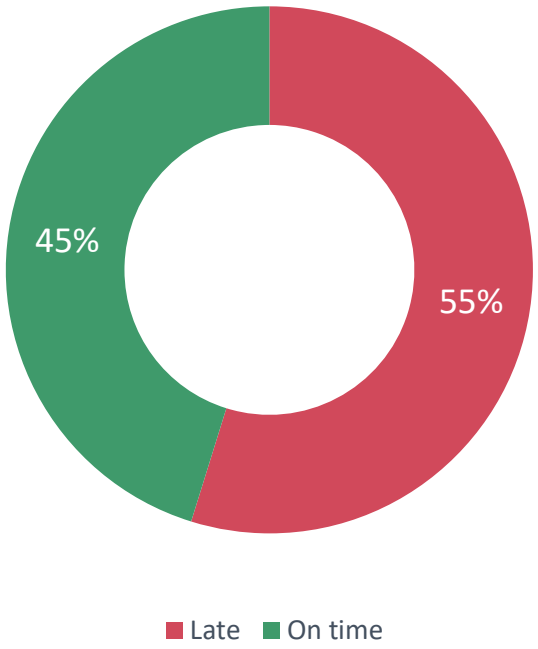
- Customer personal data removed
- Dates parsed; delay = actual − scheduled days
- Item features rolled up to each order

RESULT **One order-level table** for analysis and modelling · item detail kept for the category analysis

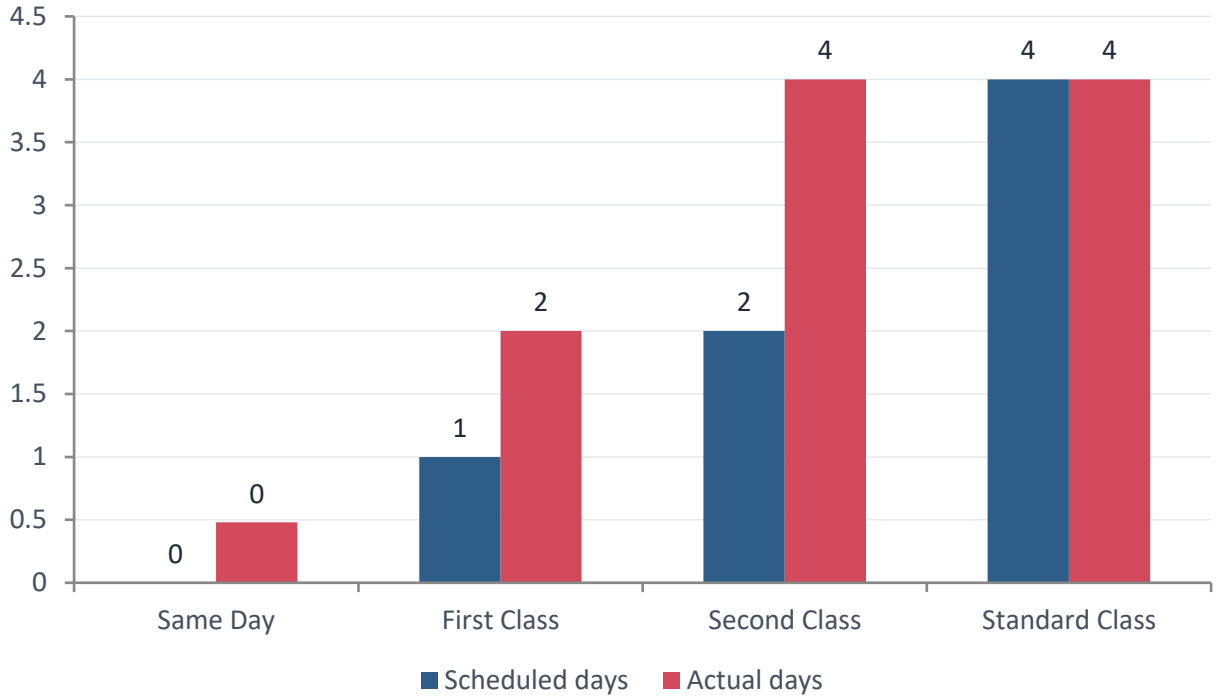
Exploratory analysis

The target is balanced, and faster modes run later than promised

Target balance (% of orders)



Average days by shipping mode



TAKEAWAY About 55% late versus 45% on time – a balanced target. First and Second Class take 1–2 days longer than scheduled.

Objective 1: Shipping mode drives lateness

↩ Back to objectives

Late Delivery
RISK PREDICTION

% of orders delivered late by shipping mode



What the data shows

- First Class: 95.3% late (Second Class 76.7%)
- Standard Class: 38.1% late
- Promised → actual days: First 1→2, Second 2→4, Standard 4→4
- Chi-square p ≈ 0, Cramér's V = 0.456 (strong)

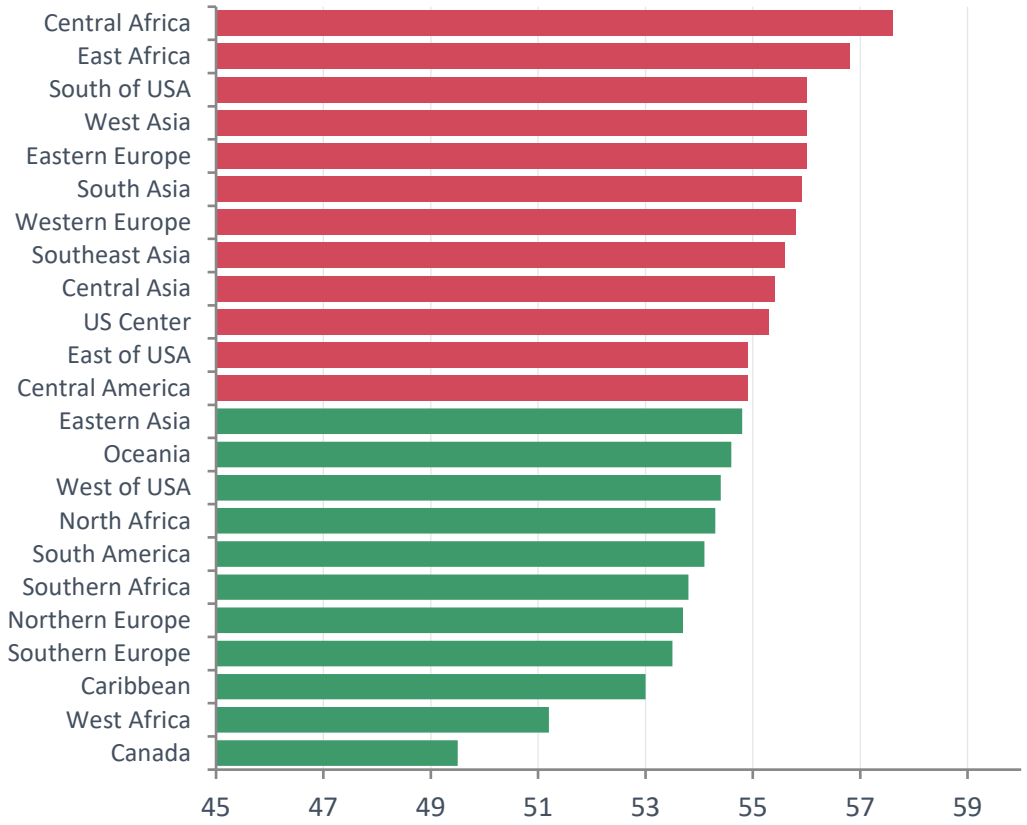
TAKEAWAY Fast modes promise 0–2 days but deliver in 1–4 – the promise is the problem, not just the carrier.

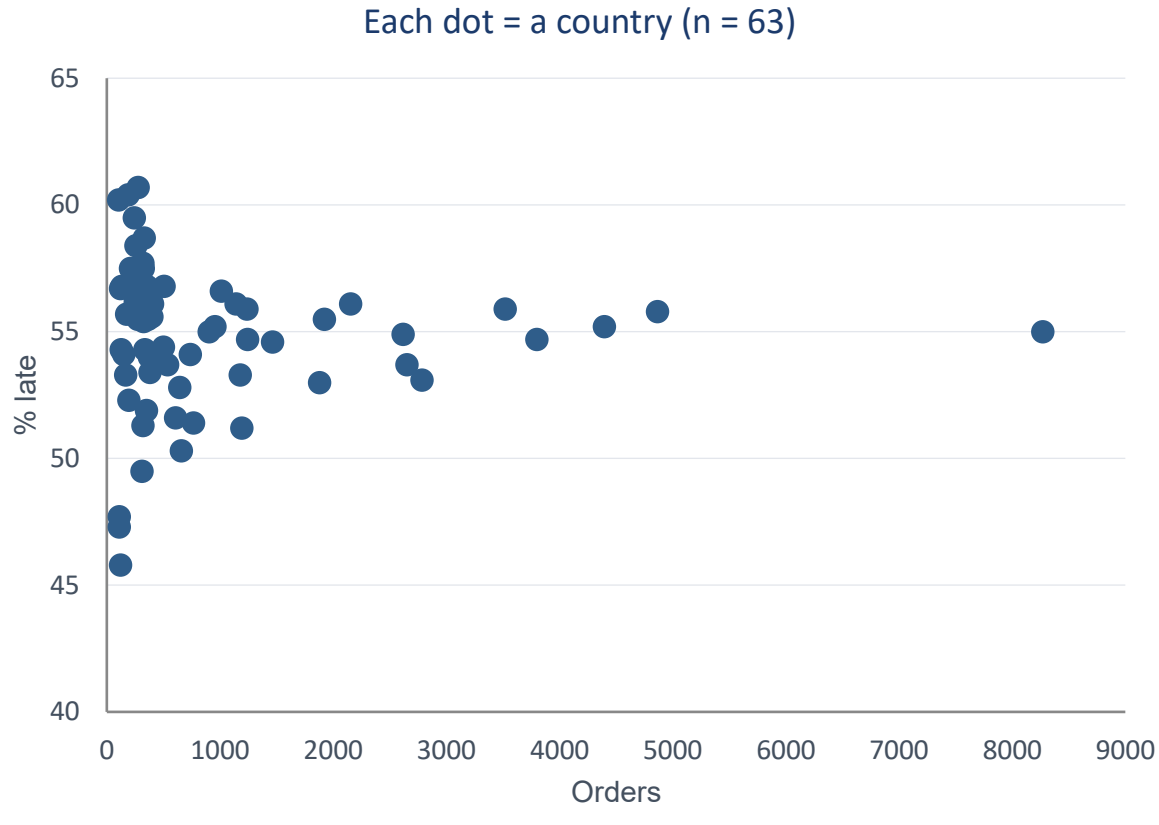
Objective 2: Geography barely matters

↩ Back to objectives

Late Delivery RISK PREDICTION

Late % by region and by country (countries with 100+ orders)





TAKEAWAY Regions span only 49.5%–57.6% (χ^2 p = 0.086, V = 0.022): not significant. Country extremes rest on small samples.

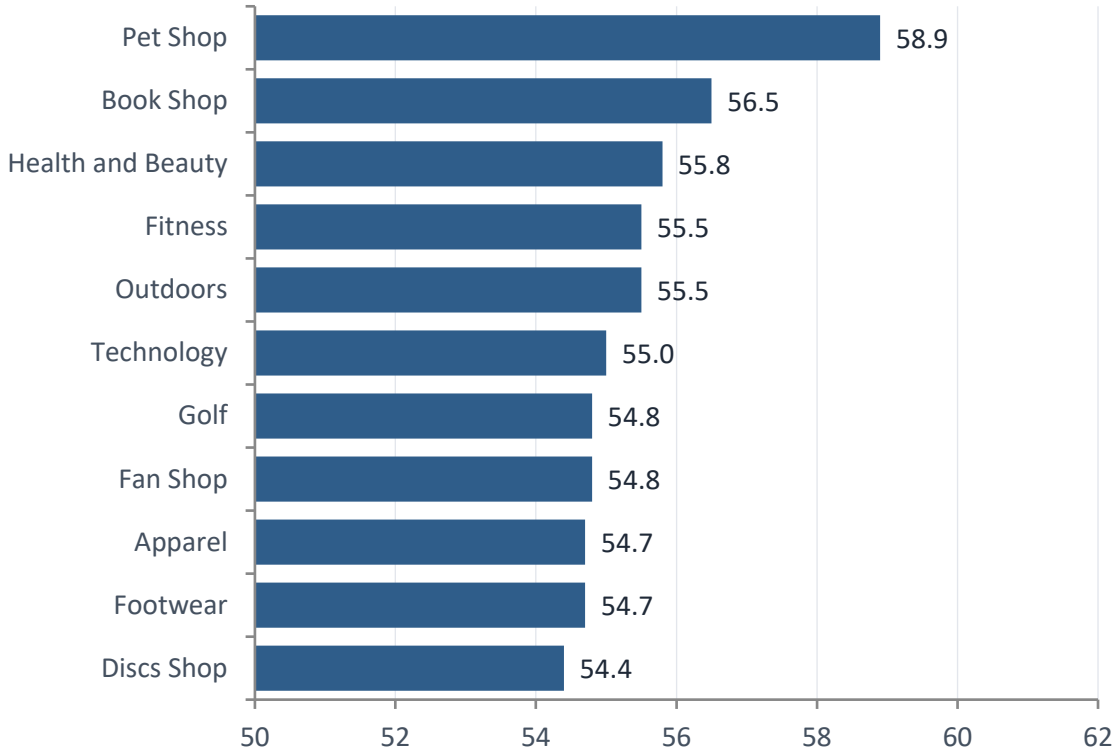
Objective 3: Product category has little effect

↩ Back to objectives

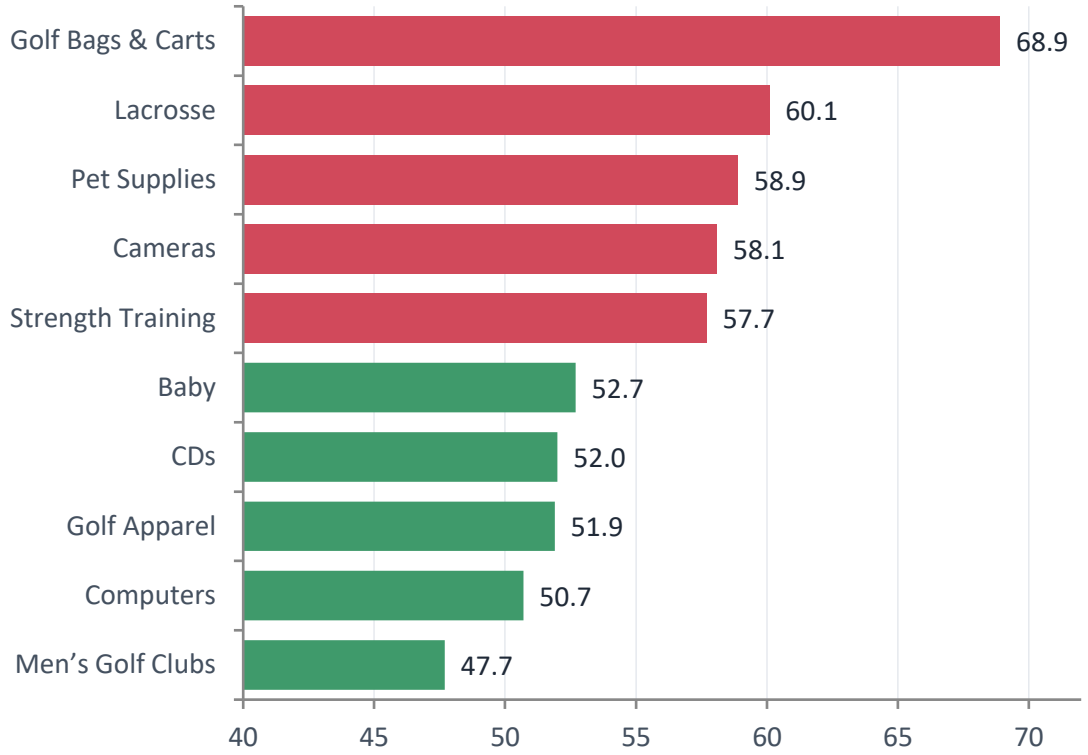
Late Delivery
RISK PREDICTION

Item level – each item inherits its order's outcome

Late % by department



Lowest and highest categories



TAKEAWAY Departments span 54.4%–58.9% (χ^2 p = 0.756, V = 0.006). Golf Bags & Carts (68.9%) has only 61 items.

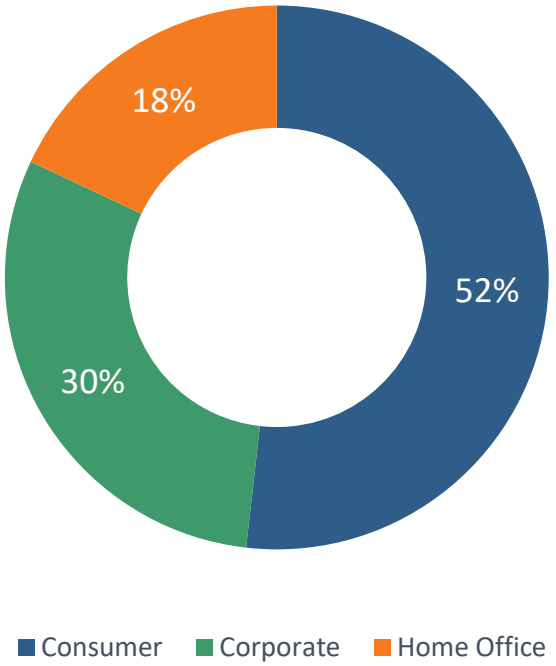
Objective 4: Every segment is equally exposed

↩ Back to objectives

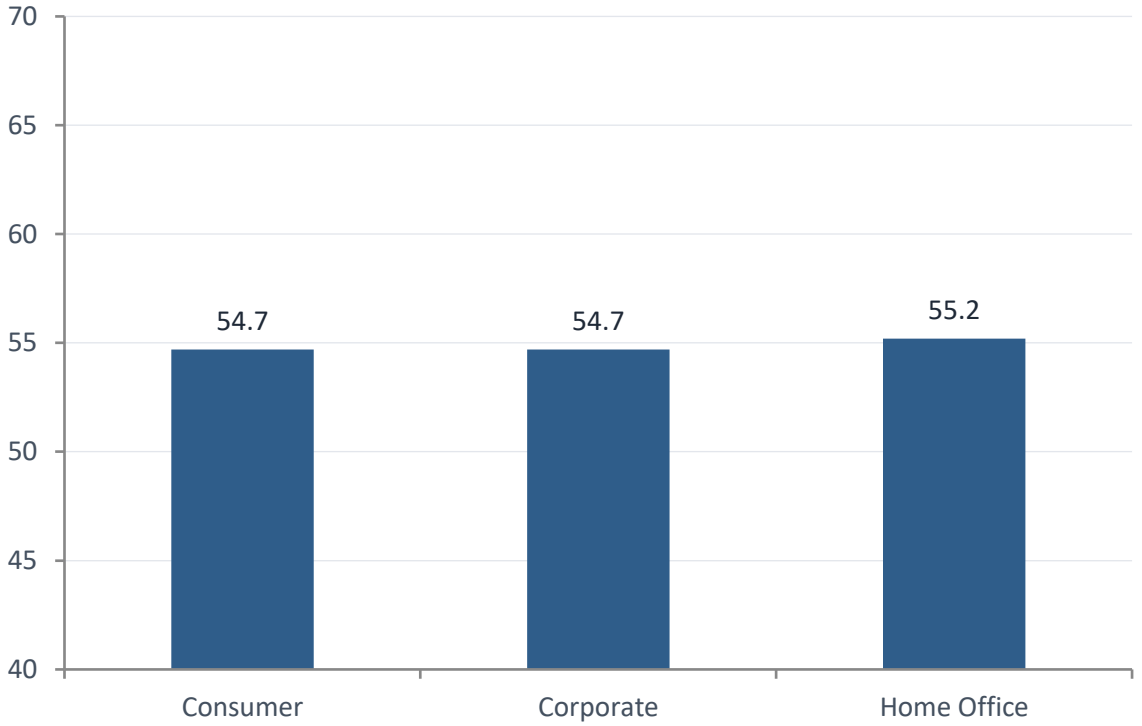
Late Delivery
RISK PREDICTION

Share of late orders mirrors segment size, not higher risk

Share of late deliveries (%)



Late % by segment (overall 54.8%)



TAKEAWAY

54.7% / 54.7% / 55.2% (χ^2 p = 0.613). Customer value tier also shows no effect (p = 0.991): no case for segment-specific policy.

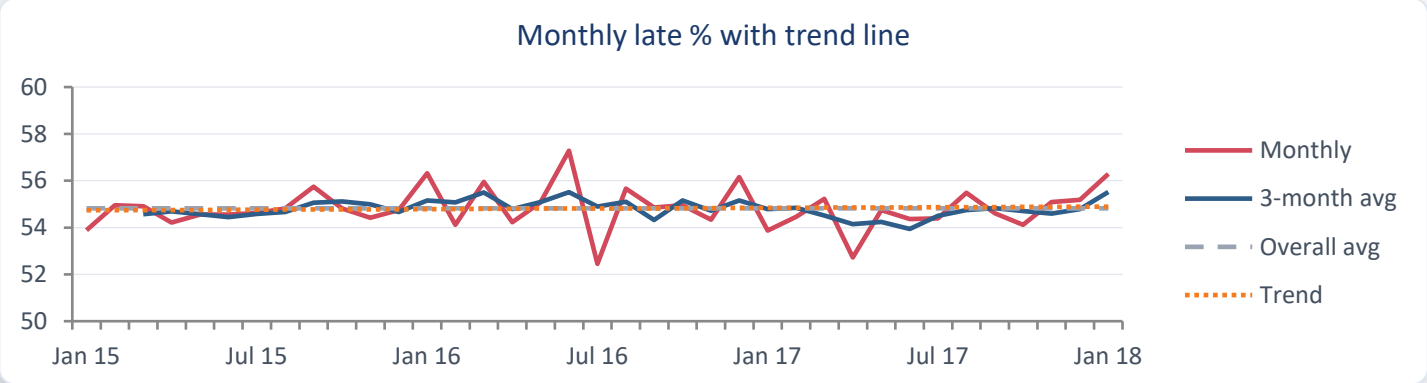
Objective 5: Temporal trends

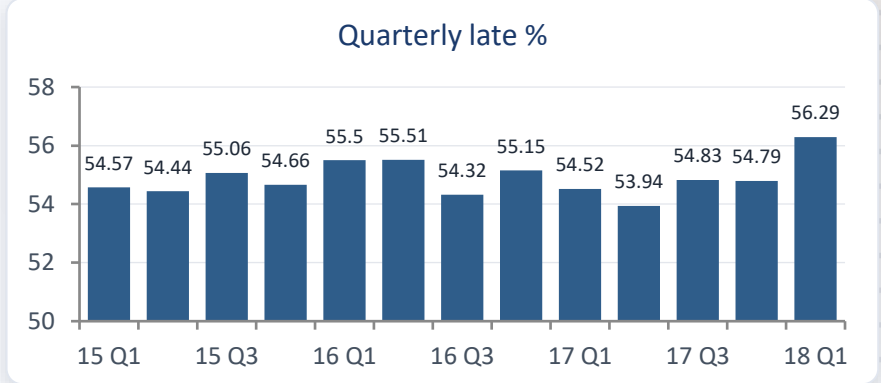
↩ Back to objectives

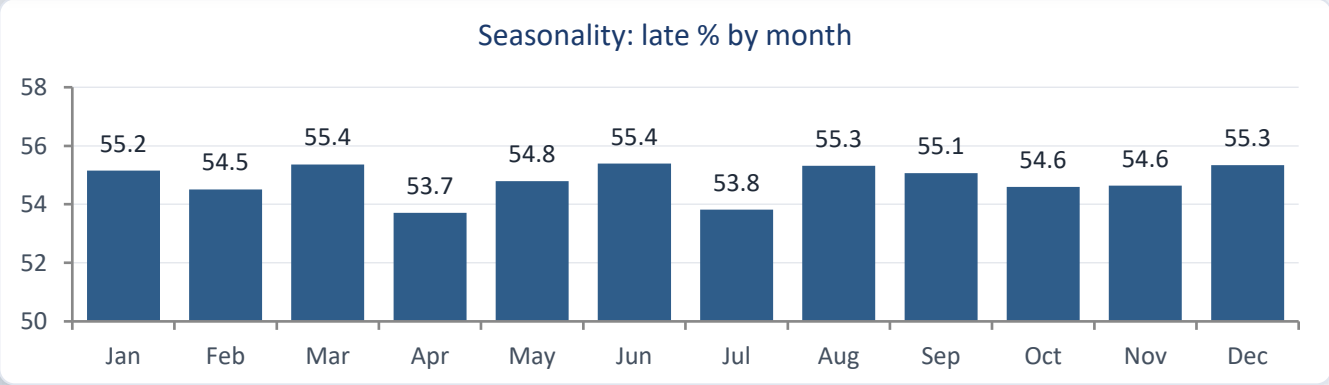
Late Delivery RISK PREDICTION

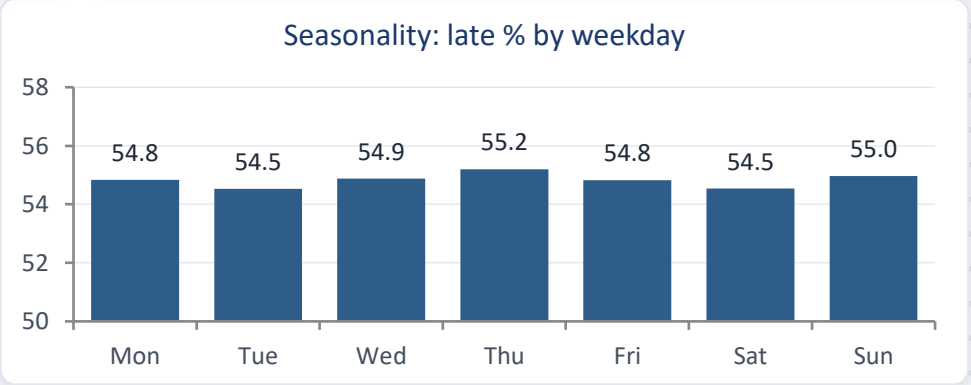
No trend and no seasonality – late % stays close to the 54.8% average

54.8%	Overall late %
56.3%	Latest month (Jan 2018)
65,752	Orders



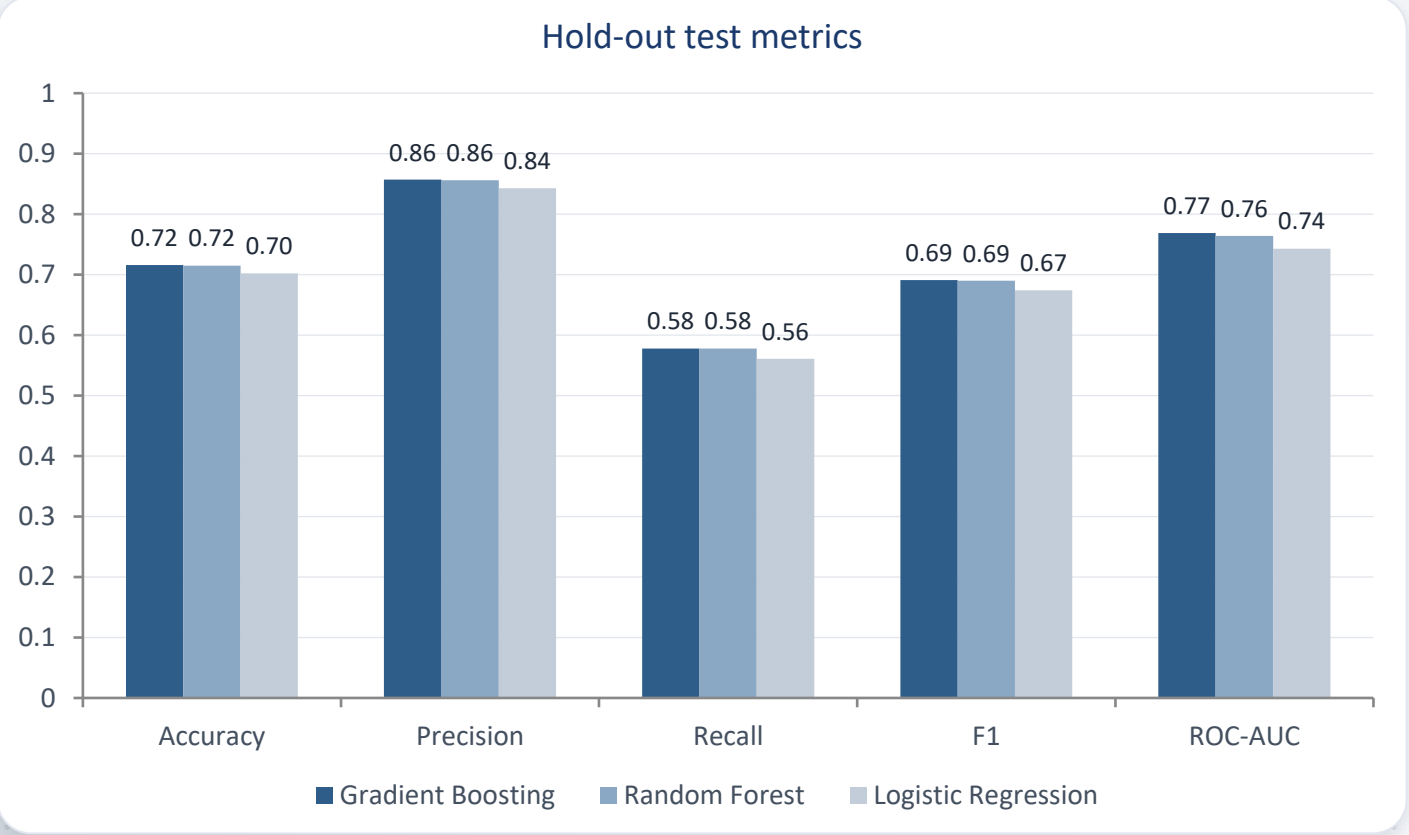






Predictive model: three models compared

80/20 stratified split; only information known at order time



Gradient Boosting – confusion matrix

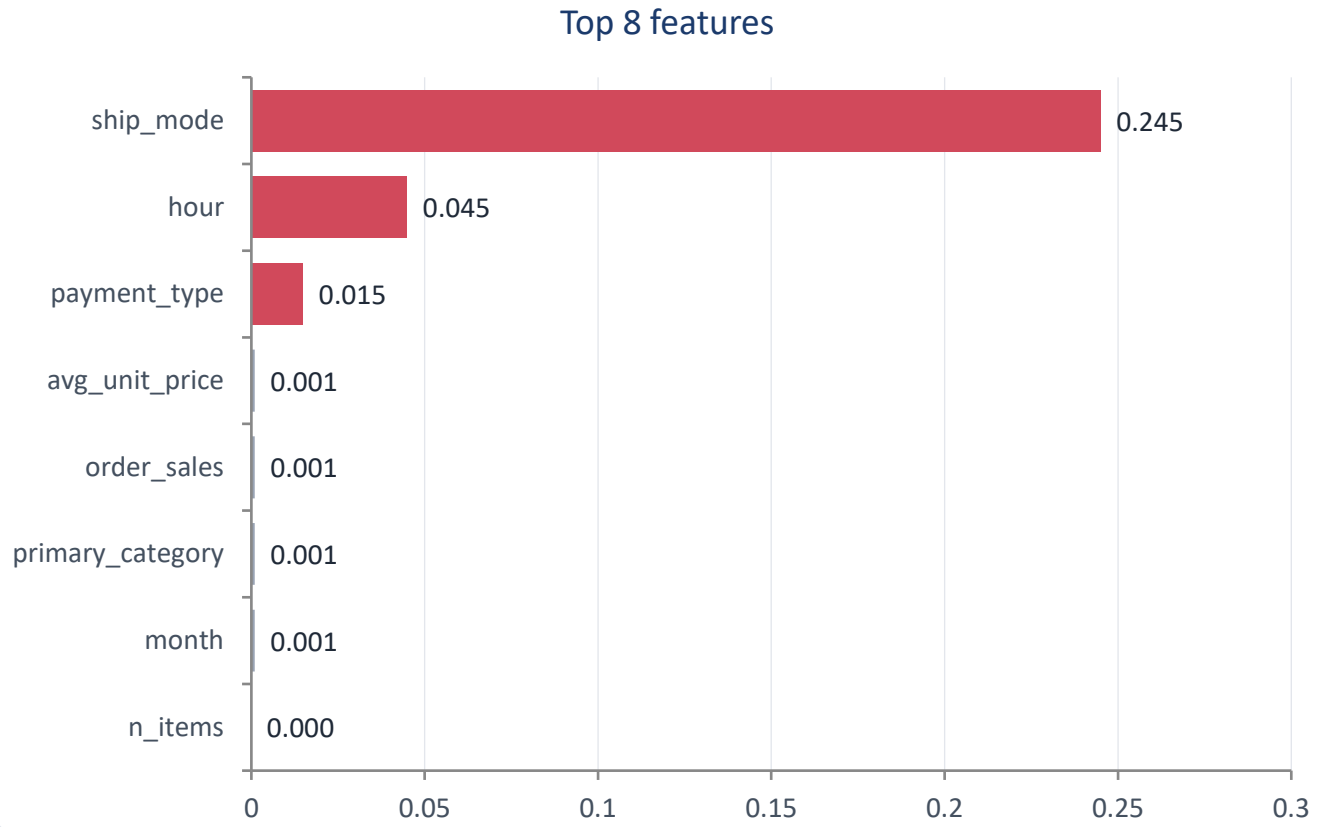
	Pred. on time	Pred. late
On time	5,243	698
Late	3,040	4,170

13,151 test orders. 5-fold CV ROC-AUC: 0.769 ± 0.004 – stable, not overfit.

TAKEAWAY **Gradient Boosting:** AUC 0.769, precision 85.7% (flags are reliable), recall 57.8% (some late orders are missed).

What drives the prediction?

Permutation importance – drop in AUC when a feature is shuffled



Reading it

- Shipping mode (≈0.245) is by far the biggest signal
- Order hour (≈0.044): a possible cut-off or processing-time effect
- Payment type (≈0.015): minor
- Region, country, category and segment ≈ 0

TAKEAWAY The model confirms the descriptive story: how we ship matters; where and what we ship barely do.

Findings and recommendations

What we found, and what to do

Key findings

- **54.8% of orders are late**
  Target is balanced
- **Shipping mode is the only strong driver**
  Cramér's V = 0.456
- **Geography, category, segment, time are flat**
  No statistically meaningful effect
- **Model reaches AUC 0.77**
  Using order-time data only

Recommendations

1. **Reset delivery promises for First/Second Class**
   Promised 1–2 days vs actual 2–4 – or add carrier capacity
2. **Score every order at checkout**
   Act on high-risk orders: notify the customer or upgrade the carrier
3. **Investigate order hour and payment type**
   Cut-offs, payment holds
4. **Avoid region- or category-specific fixes**
   No evidence they pay off
5. **Enrich the data**
   Carrier, warehouse, stock, distance – to lift AUC

LIMITS One company, 2015–Jan 2018; recall of 58% means tuning the threshold to the cost of a missed late order.

Late Delivery
RISK PREDICTION

Q&A

Questions?

Thank you for listening

Late Delivery Risk Prediction · DataCo supply chain capstone

54.8%
of orders arrive late

V 0.46
shipping mode is the one strong driver

AUC 0.77
model flags risk at checkout



TRY IT Live model: **late-delivery-risk-prediction-dataco.streamlit.app**

Edwin Tan